

Omnidirectional and Equirectangular ChArUco Calibration

Detailed Experimental Report

Prepared for project progress documentation

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Summary

This report documents the calibration work performed on a wide-angle/omnidirectional camera using a ChArUco board. The main work included validating the existing two-stage ChArUco pipeline, implementing OpenCV omnidirectional calibration, testing true longitude-latitude/equirectangular rectification, and analyzing why the generated equirectangular images were not suitable for robust ChArUco detection on the current dataset.

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1. Main objective

The main objective of this work was to investigate whether a preliminary calibration of a wide-angle or omnidirectional camera can be used to generate rectified or equirectangular views that make ChArUco marker detection more reliable. The long-term goal is to improve calibration quality by recovering more useful ChArUco points from difficult frames. The dataset is challenging because the board is very close to the camera, strongly distorted, partially visible in many frames, and sometimes located near the edge or seam of the camera projection.

The work was divided into two parts. First, the existing two-stage ChArUco calibration pipeline was verified and used as a baseline. Second, a true omnidirectional calibration was implemented using OpenCV omnidir, followed by experiments with longitude-latitude/equirectangular rectification. The equirectangular part was tested carefully because it was the main projection type of interest, but the outputs were also evaluated visually to decide whether they were actually useful for detection.

2. Input data and calibration target

The input is a calibration video recorded with a wide-angle or omnidirectional camera. The calibration target is a ChArUco board. ChArUco boards combine ArUco markers with chessboard-like inner corners, so the marker IDs help identify the board and the interpolated ChArUco corners provide precise calibration points.

The raw video frames show heavy geometric distortion. The board is often very close to the lens, which causes strong curvature and stretching. In several frames the board is partly cut off, and some markers are blurred or close to the image boundary. These properties make the dataset difficult for both normal pinhole calibration and equirectangular unwrapping.

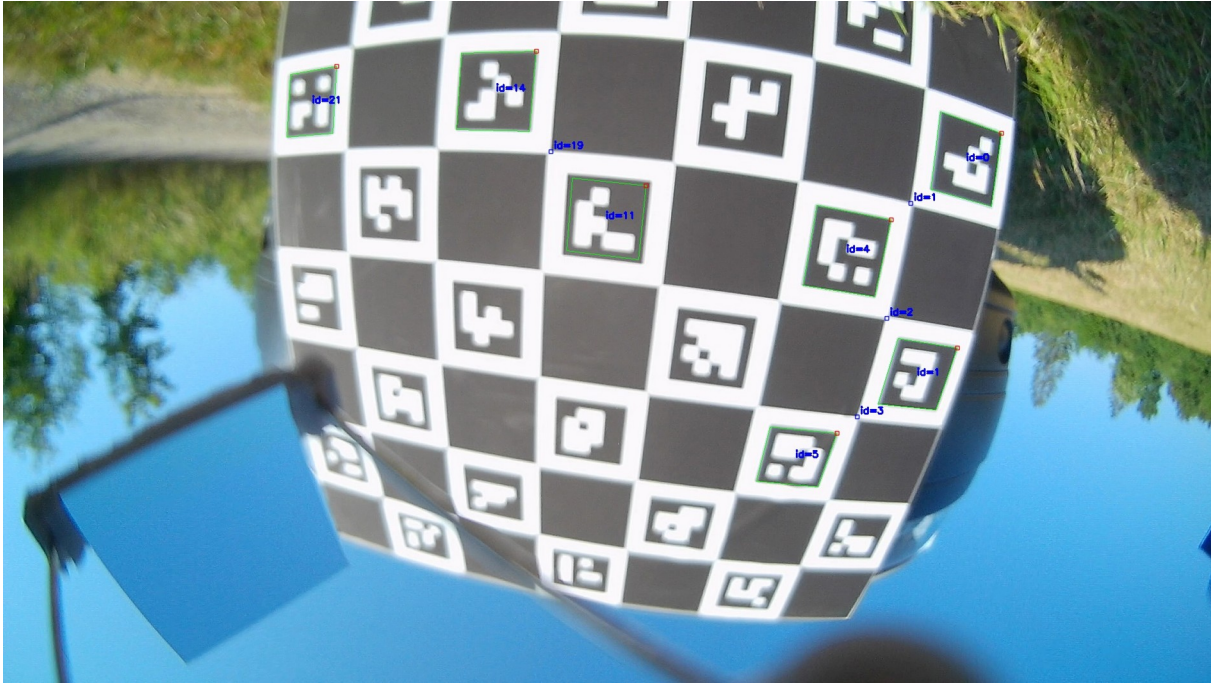


Figure 1. Example raw ChArUco detection. The detected marker outlines and ChArUco IDs show that raw detection works, but the board is strongly distorted and close to the camera.

3. Baseline two-stage ChArUco calibration pipeline

The first working pipeline is a two-stage ChArUco calibration pipeline. It begins by sampling frames from the calibration video and detecting ArUco markers and ChArUco corners in the raw distorted images. From 372 sampled frames, 274 frames contained detectable ArUco markers, and 67 frames were marked as useful for calibration. Visual inspection confirmed that the detections were on the real calibration board rather than on random background regions.

A preliminary calibration was then performed using OpenCV standard ChArUco calibration. This is a pinhole camera calibration with radial and tangential distortion coefficients. From the useful raw frames, 19 frames had enough ChArUco corners for the stricter preliminary calibration step. This preliminary calibration produced an RMS reprojection error of 1.893938 pixels.

The preliminary calibration was used to rectify the useful raw frames. The purpose of this intermediate rectification was not to produce the final camera model, but to make the board easier to detect in a second pass. All 67 useful raw frames were successfully rectified. The rectified images contained black invalid regions and stretching, but the board was often easier to interpret than in the raw omnidirectional view. ChArUco detection was then repeated on the rectified frames, and 35 frames were considered useful for final calibration.

The final calibration using the rectified detections produced an RMS reprojection error of 0.869140 pixels. This is a substantial improvement over the preliminary value. The relative reduction in reprojection error was approximately 54.11%. This showed that the two-stage detection and rectification pipeline was useful as a baseline.

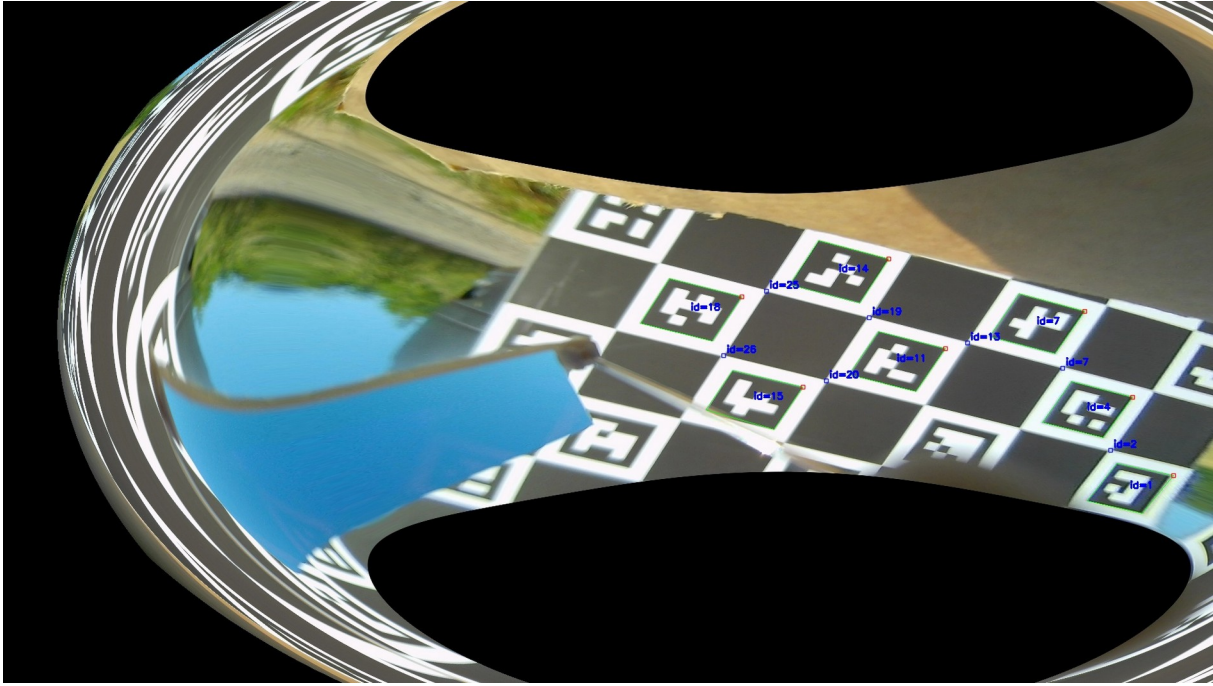


Figure 2. Example second-pass detection after rectification. The image has black invalid areas, but many markers and ChArUco corners are visible.

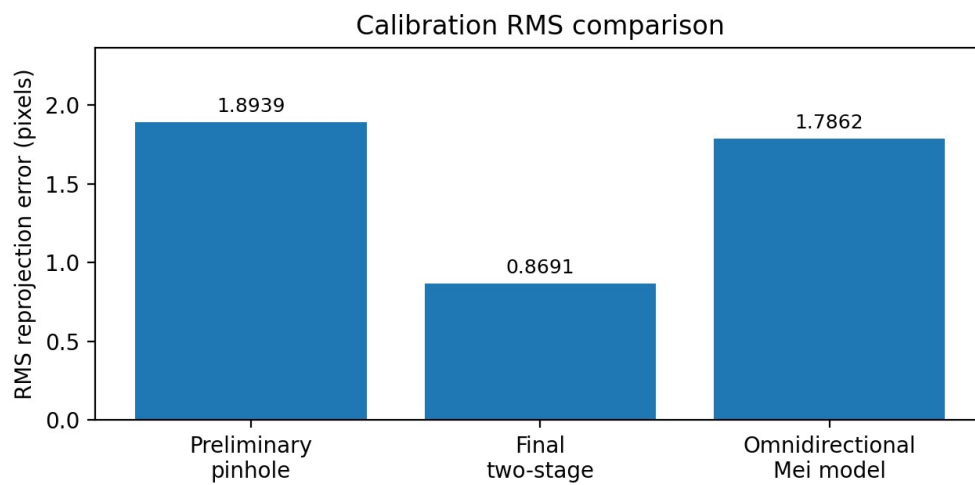


Figure 3. RMS reprojection error comparison for the main calibration results.

The main numerical results are summarized below:

Stage	Model / method	Frames	RMS error
Preliminary calibration	Standard ChArUco / pinhole	19	1.893938 px
Final calibration	Two-stage rectified ChArUco / pinhole	35	0.869140 px
Omnidirectional calibration	OpenCV omnidir / Mei model	19 passed; 11 retained internally	1.786235 px

4. Omnidirectional calibration with OpenCV omnidir

After verifying the baseline pipeline, a true omnidirectional calibration was implemented using OpenCV omnidir. This is different from the standard pinhole calibration because it estimates an additional omnidirectional model parameter, ξ . The calibration uses the same ChArUco point correspondences from the raw frames, but the camera model is more suitable for wide-angle or omnidirectional projection.

The OpenCV omnidirectional calibration script collected 3D ChArUco board coordinates and their matching 2D image coordinates. From the 67 useful raw frames, 19 frames had enough ChArUco corners to be passed to `cv2.omnidir.calibrate`. OpenCV internally retained 11 frames according to the returned `idx` output. The omnidirectional calibration produced an RMS reprojection error of 1.786235 pixels and estimated $\xi = 0.80027956$.

This result is important because it confirms that the project now has a true omnidirectional camera model, not only a standard pinhole model. The output parameters are K , ξ , D , $rvecs$, $tvecs$, and `idx`. These parameters are required for OpenCV omnidirectional rectification functions such as `RECTIFY_LONGLATI` and `RECTIFY_PERSPECTIVE`.

Parameter	Estimated value / description
f_x, f_y	approximately 1402.16, 1405.16 pixels
c_x, c_y	approximately 994.11, 527.28 pixels
ξ	0.80027956
distortion D	$[-0.29508521, 0.05461258, 0.00557020, -0.00819621]$
RMS	1.786235 pixels

5. Equirectangular rectification experiments

The next experiment was to generate equirectangular or longitude-latitude images using the omnidirectional calibration result. This was implemented with OpenCV omnidir `initUndistortRectifyMap` and the `RECTIFY_LONGLATI` flag. Technically, this step worked: all 67 useful raw frames were converted into equirectangular outputs and saved successfully.

However, visual inspection showed that the first equirectangular outputs were not suitable for ChArUco detection. The images contained very large black regions, and the useful scene appeared only as separated curved islands. The ChArUco board was often split across different parts of the output or compressed near the boundary. This happened because the camera does not provide a full clean spherical view and because the board is very close to the omnidirectional camera.



Figure 4. Initial equirectangular output using RECTIFY_LONGLATI. The transformation runs correctly, but the valid image region is fragmented and dominated by black areas.

Several adjustments were tested to improve the equirectangular output. The output width was increased to create a wider 2:1-like canvas. Rotated equirectangular views were also tested using different yaw and pitch values. Finally, local equirectangular windows with reduced angular ranges were tested. These experiments were intended to center the ChArUco board and reduce the amount of invalid black area.

The experiments showed that simply reducing the angular range does not solve the problem. Without a good spherical viewpoint and a board that stays inside the valid region, the local longitude-latitude images become even more distorted. In several tests, the board became stretched into thin vertical strips, split at the seam, or moved almost completely outside the valid region. Therefore, the equirectangular approach was implemented successfully, but it did not create detection-friendly frames for this dataset.

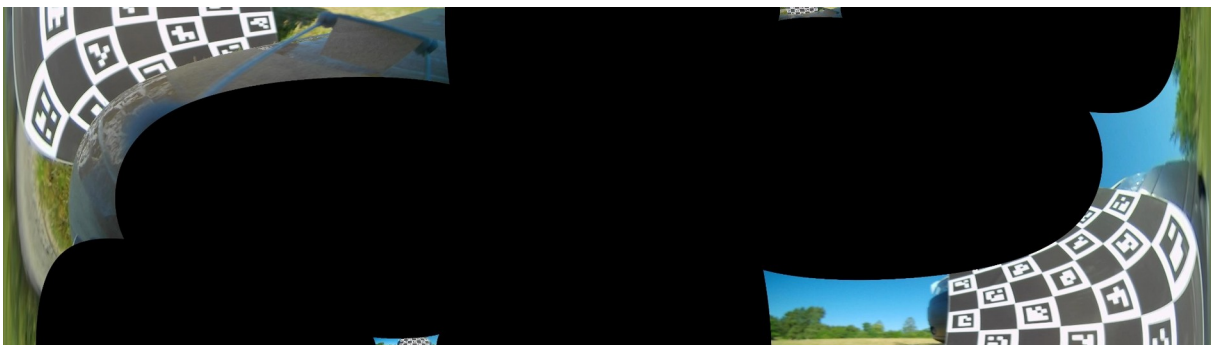


Figure 5. Rotated equirectangular test. The board is visible, but it is still split and distorted by the longitude-latitude mapping.



Figure 6. Local reduced-angle equirectangular test. The output is more focused, but the board becomes heavily stretched and not suitable for reliable detection.

6. Perspective rectification sanity check

Because the equirectangular results were not useful for detection, perspective rectification was tested as a sanity check. This step still uses the same omnidirectional calibration parameters, but it uses RECTIFY_PERSPECTIVE instead of RECTIFY_LONGLATI. The goal was not to replace the equirectangular experiment, but to check whether a virtual perspective view could produce more normal-looking images of the board.

The perspective rectification produced clearer views than the full equirectangular output. Multiple yaw and pitch views were generated, including front, left, right, up, down, and additional intermediate angles. In several views, the ChArUco board was visible and less fragmented. However, many images were still too zoomed in, partially cut off, or blurred. Some perspective views were therefore better for visual inspection, but they were still not clearly superior to the original two-stage rectification baseline.

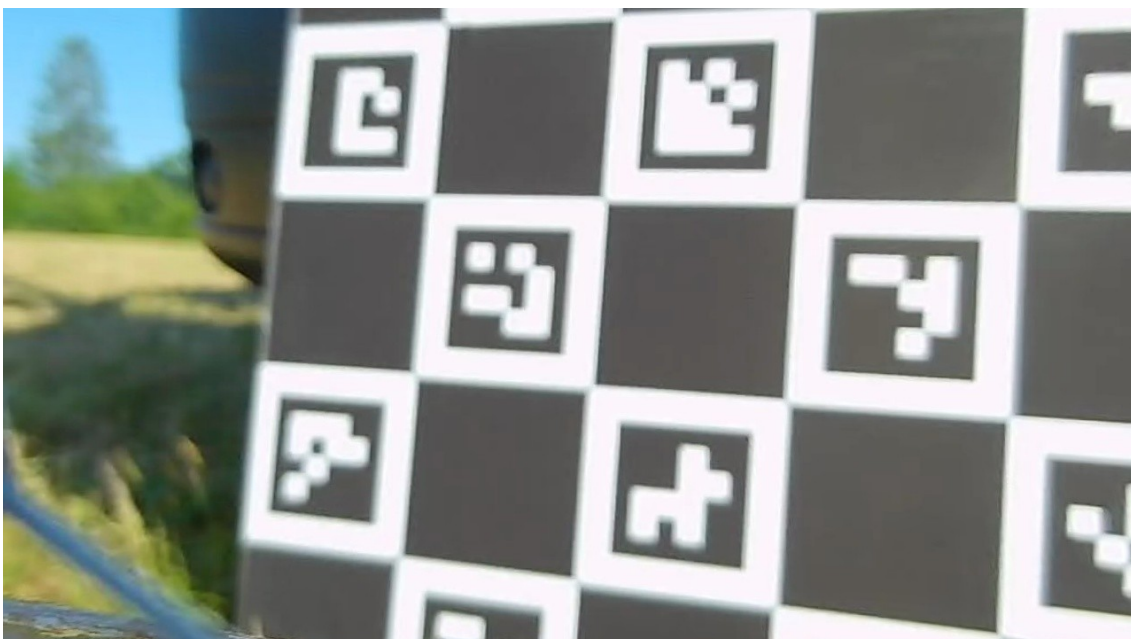


Figure 7. Example perspective rectified view. This is not equirectangular; it is a virtual perspective view generated from the omnidirectional model.

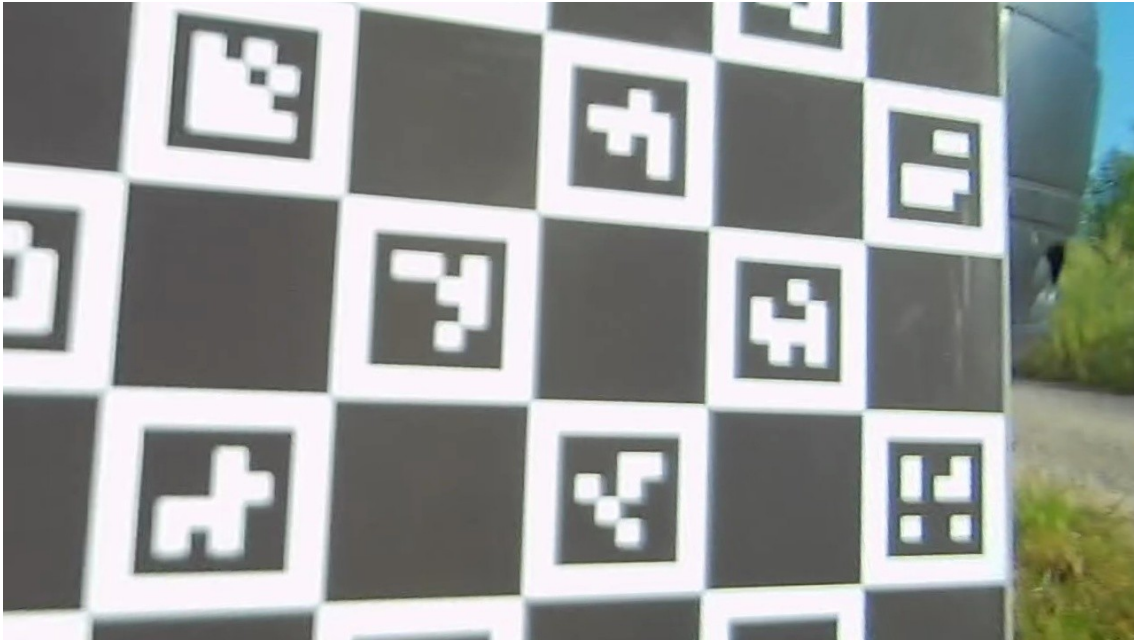


Figure 8. Another perspective rectified view. The board is more locally visible, but it remains cropped and not ideal for robust detection.

7. Discussion and interpretation

The experiments show that the main difficulty is not only the projection model, but also the geometry of the recorded data. The calibration board is too close to the camera and often occupies extreme regions of the omnidirectional image. Equirectangular projection is useful when a scene can be represented as a reasonably continuous spherical panorama. In this dataset, the board crosses difficult regions of the projection and gets split or strongly stretched when mapped into longitude-latitude coordinates.

The equirectangular transformation should therefore be interpreted as an implemented and tested experimental step, not as the final detection strategy. It proves that the omnidirectional calibration parameters can be used with RECTIFY_LONGLATI, but it also shows that this projection is not appropriate for recovering ChArUco detections in the current video. This is a useful negative result because it prevents spending more time forcing an unsuitable projection.

The strongest quantitative improvement still comes from the two-stage ChArUco pipeline. The second-pass detection after rectification increased the number of useful calibration frames from 19 to 35 and reduced RMS error from 1.893938 pixels to 0.869140 pixels. This makes the two-stage pipeline the best reliable calibration result so far.

8. Final conclusion and next steps

This work successfully implemented and evaluated three related calibration paths. The first path is the existing two-stage ChArUco pipeline with intermediate rectification, which gave the best practical result. The second path is true omnidirectional calibration using OpenCV omnidir, which successfully estimated the ξ parameter and produced a valid omnidirectional model. The third path is equirectangular rectification using RECTIFY_LONGLATI, which technically worked but did not produce useful detection frames on this dataset.

The final conclusion is that equirectangular rectification is not currently useful for improving ChArUco detection in this video because the board is too close, partially visible, and strongly distorted. The most reliable result remains the two-stage rectification-based ChArUco pipeline. The omnidirectional model is still valuable because it provides a correct camera model and can be used for further experiments, but the equirectangular images should not be used as the main detection input unless the recording setup is improved.

The next recommended step is to collect a new calibration video with better spatial coverage. The board should appear in more regions of the image, especially the poorly covered side regions, and it should be farther from the camera so that the markers are less curved and less likely to cross projection seams. If a new video is not possible, the current best result should be documented as the two-stage ChArUco calibration, with the equirectangular experiment included as an analysis of a tested but unsuitable approach for this dataset.

Short update paragraph for progress report

Today I verified the existing two-stage ChArUco calibration pipeline and then implemented a true omnidirectional calibration using OpenCV omnidir. The two-stage baseline worked well and improved the reprojection error from 1.8939 px to 0.8691 px. The omnidirectional calibration also ran successfully and estimated $\xi = 0.8003$ with an RMS error of 1.7862 px. I then tested true equirectangular rectification with RECTIFY_LONGLATI, including rotated and reduced-angle variants. Although the transformation technically worked, the resulting images were not suitable for ChArUco detection because the board was split or heavily stretched in longitude-latitude space. Therefore, the best reliable result remains the two-stage ChArUco pipeline, while the equirectangular experiment is documented as a tested limitation of the current dataset.

References

OpenCV documentation: cv::omnidir namespace, omnidirectional calibration and rectification functions, including RECTIFY_LONGLATI and RECTIFY_PERSPECTIVE.

https://docs.opencv.org/4.x/db/dd2/namespacencv_1_1omnidir.html

OpenCV ccalib omnidir header documentation: initUndistortRectifyMap and supported rectification flags.

https://docs.opencv.org/4.x/db/d62/omnidir_8hpp.html

Project outputs generated during the current experiment: raw detection summaries, preliminary and final calibration files, omnidirectional calibration output, equirectangular frames, and perspective rectification tests.